Humusmus

sounds that grow into each other



USER GUIDE · VERSION 1.0

K.ODK

Sounds that grow into each other

Humusmus is a small garden of sounds. Up to eight recordings stand on it as flowers, beside two strange plants that make data and tone of their own. Every sound is listened to while it plays, and what one flower does can move another — so the recordings start to answer each other, and the piece keeps changing by itself.

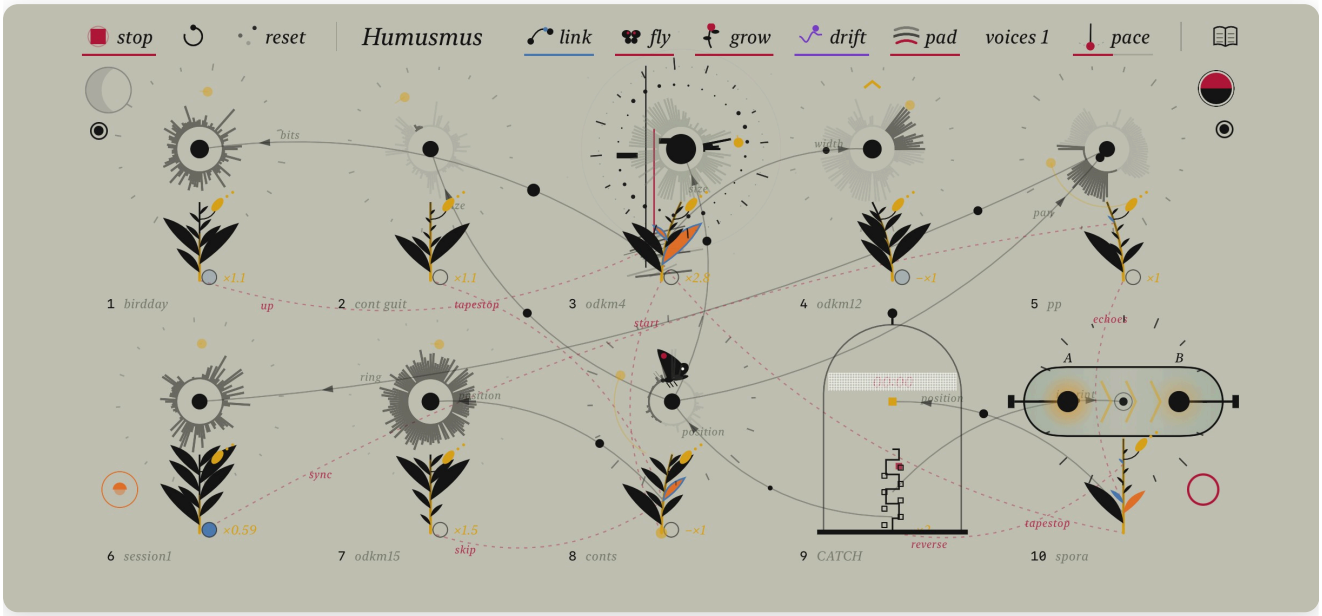
You do not arrange a song here. You plant sounds, tie them together, and then listen and lean in: touch a flower, turn on a key, let the camera look. It is meant to be played like an instrument, for listening and improvising, more than for building a finished track.

> TOO MUCH TO LEARN? SKIP IT.
>
> 1. Tap an empty flower and load two or three sounds — field recordings work well. Or record straight into them with `mic`.
> 2. Press `play`, then turn on `link` and `drift`.
> 3. Tap the *moon* (top left) and point the phone at something that moves.
>
> That is already the whole instrument. Everything below is for when you want to steer it.

Contents

The plate	what you see on the main screen
Flowers 1 – 8	your sounds, and a flower's panel
Links and gates	how one sound moves another
Catch and Spora	the jar and the tube
Who is heard	voices, hold, the sun, pad, the moon
The bar	every key along the top
Now and then	bursts, lasers, shots, keep and take
Requirements	devices, camera, microphone, versions

The plate



The plate while playing: eight sound flowers, the jar (9) and the tube (10), tied by links and gates.

Everything happens on one screen, the *plate*. The bar along the top holds the keys; the rest is the garden.

Flowers 1 – 8	Your sounds. Each one is read as it plays: its eight bands, from 63 Hz to 8 kHz, are its leaves. An empty place shows a dotted circle with a +.
9 · the jar	*CATCH* — never heard, only read. It walks over the mix and turns what it finds into data for the others.
10 · the tube	*SPORA* — two oscillators and a two-peak filter. Heard always, now and then, or only as data.
Moon	Top left. Tap it and the camera looks; the picture ties and lets in the sounds.
Sun	Top right. Decides how often the loud and the quiet moments come through.
Small rings	Under the moon and the sun: *KEEP*, two memories of the whole plate.
Orange sun	Bottom left: where the pad listens (its band).
Red ring	Bottom right: *TAKE*, records everything you hear to a file.

Lines on the plate

A *link* is a fine ink line from the leaf it reads to the flower it moves; a seed runs along it, as big as the data, and what it moves is written beside it. A *gate* is a red dashed line between two feet, lit when it fires. Spora's sends are red dotted lines.

A stem coloured from its foot upward is a flower being moved right now — the higher the colour, the harder the push. Tap the name *Humusmus* in the bar to switch to a lighter drawing (dots instead of letters, same motion); tap again to go back.

Flowers 1 – 8

Planting a sound

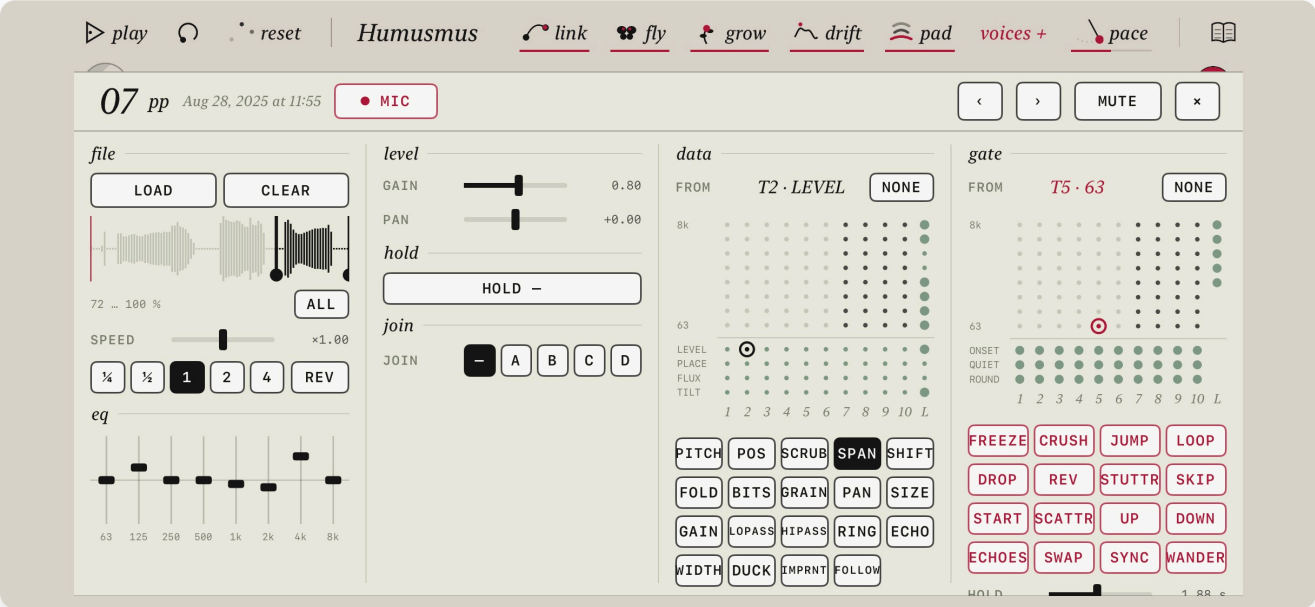
Tap an empty flower to load a file. Or use `mic` in the bar: it records into flowers 1, 2, 3 … 8 in turn (only while stopped). Each flower also has its own `• MIC` in its panel, beside its date.

Touching a flower

Tap a full flower to open its panel. *Hold* a flower to fire a laser from it at once (see *Now and then*).

Touch its leaves to hear its sound right now. Slide up or down to bend its pitch, up to an octave either way; let go and it returns. Two flowers can be held at once.

The small figure under a flower shows its speed, for example `×1.5` or `−×1` when it plays backwards.



A flower's detail panel: file, level, hold and join, then its data link and its gate.

A flower's panel

The first panel is drawn with moving pictures — drag them: *gain*, *pan*, *speed* (turn it past zero to play backwards), *hold* and *join*, with the *eq* and *tape* below. `DETAIL` opens the full panel shown above.

file	`LOAD` another file, `CLEAR` it. Drag the ends of the waveform to choose the part that plays; `ALL` resets. *Speed* with steps ¼ ½ 1 2 4 and `REV`. An eight-band *eq* below.
level	Gain and pan.
hold	Makes this the one sound that always stays under everything (see *Who is heard*).
join	Puts the flower in a group (A – D) whose members turn together.
data · gate	Its link and its gate — the next chapter.

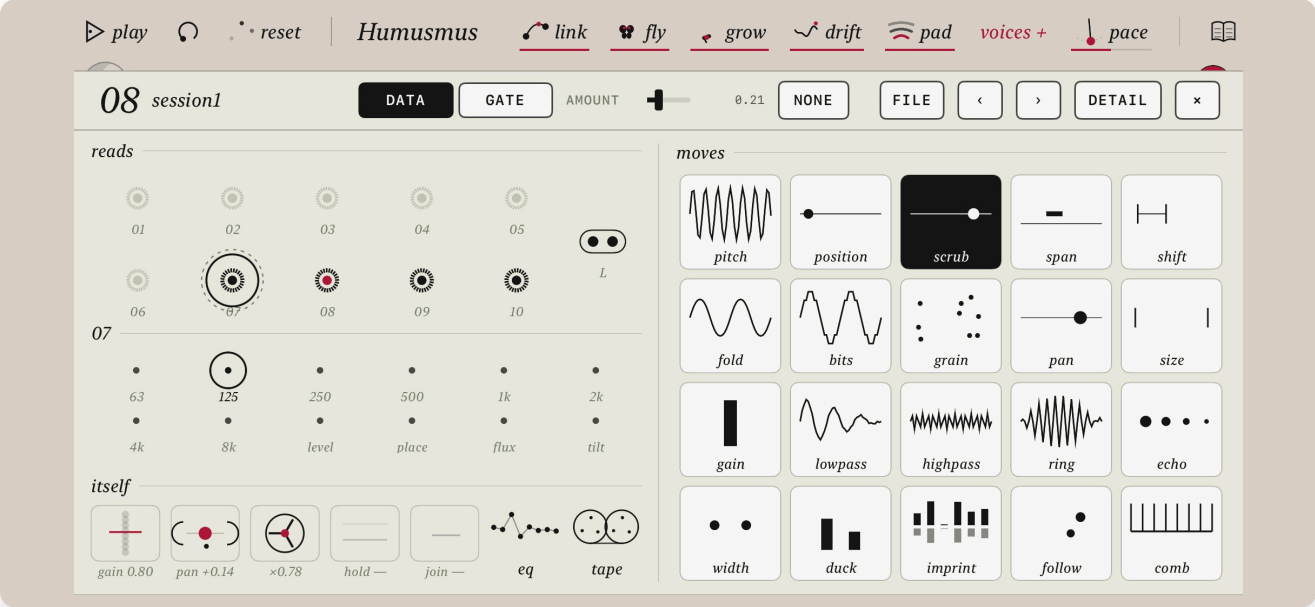
tape	Puts its sound on tape: wow, flutter, a worn top end, the head lifting now and then. Hooked to another flower, it is pulled as that one swells; its hiss hangs over the plate as a fine fog.

In the header: ‹ › step to the neighbouring flower, MUTE silences it, × closes.

03 · HOW THEY TIE

Links and gates

Every sound is also data: how loud each of its bands is, how it rises, how it changes. There are two ways to use that data on another flower.



DATA: what to read (left), what it moves (right), and AMOUNT.

Data → moves

A *link* reads one value and sends it, continuously, to a flower. Under *reads*, pick a source flower (01 – 10, or *L* for the links' own oscillators), then one of its bands (63 Hz … 8 kHz) or one of its measures — *level*, *place*, *flux*, *tilt*. Under *moves*, pick the destination:

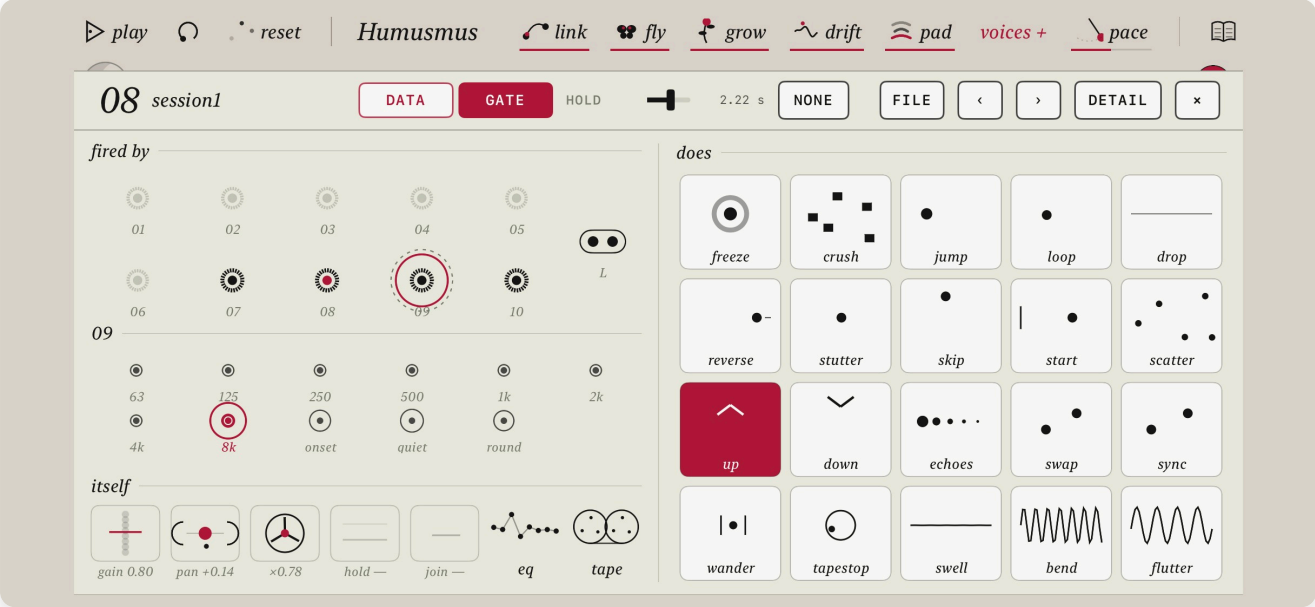
`pitch` `position` `scrub` `span` `shift` `fold` `bits` `grain` `pan` `size` `gain` `lowpass` `highpass` `ring` `echo` `width` `duck` `imprint` `follow` `comb`

AMOUNT sets how hard it pushes. NONE removes the link.

Gates → shots

A *gate* waits for a moment in the data — a band standing still, a sudden rise — and then fires a shot on the flower. Choose what fires it under *fired by* (a band, or *onset*, *quiet*, *round*), and what happens under *does*:

`freeze` `crush` `jump` `loop` `drop` `reverse` `stutter` `skip` `start` `scatter` `up` `down` `echoes` `swap` `sync` `wander` `tapestop` `swell` `bend` `flutter`



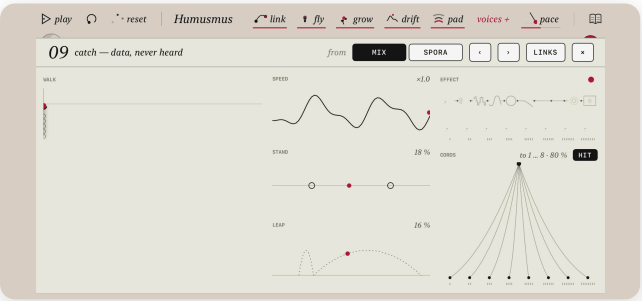
GATE: a moment in one sound fires an action on another, held for HOLD seconds.

`HOLD` sets how long the shot lasts.

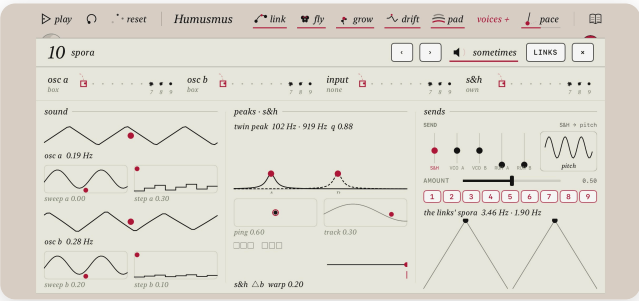
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04 · THE TWO PLANTS

Catch and Spora



CATCH (9): walk, speed, stand, leap; its cords set 1 – 8.



SPORA (10): oscillators, peaks and S&H, sends.

9 · Catch — data, never heard

The jar never makes a sound. It takes a drunk's walk over the mix (or over Spora — choose with `MIX` / `SPORA`) and turns what it hears into data for the other flowers. In its panel you see the *walk*, its *speed*, how often it *stands* and *leaps*, an *effect* chain, and the *cords*: eight strings that let it set the levels of flowers 1 – 8 (`HIT` makes it strike).

10 · Spora — the tube

Spora is a small synthesizer: two slow oscillators, *osc a* and *osc b*, feeding a filter with two resonant peaks and a sample-and-hold, all tangled together. Drag each oscillator up or down; under each are its *sweep* and *step*. The header chooses how it is heard — always, *sometimes*, or only as data.

Its *sends* let Spora move the others: choose a source (S&H, VCO A/B, RUN A/B), a destination such as *pitch*, the `AMOUNT`, and which flowers (1 – 9) receive it. The row along the top chooses what

drives each part — its own box, a flower's sound, or nothing.

05 · WHO IS HEARD

Voices, the sun, the pad, the moon

Voices

Only one to three flowers are heard at once (`voices` in the bar), chosen by the bands Spora points at; a burst breaking in counts as one. The others keep playing unheard, and are still read as data.

Hold

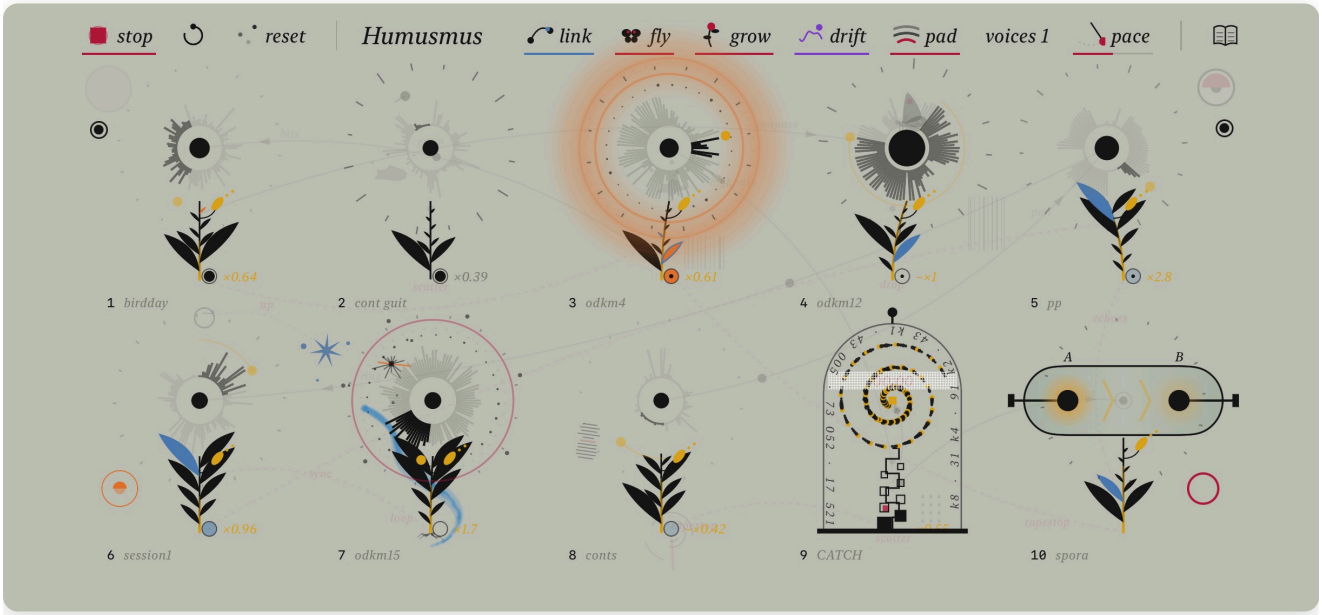
One sound, chosen in its panel, stays under all of it, always — left as it is.

The sun

Top right. Drag its upper half to set how often the *loud* moments come through; its lower half, the *quiet* ones. Two taps: everything through. Hold it to open its board.

The moon

Top left. Tap it and the camera starts to look; each flower has its own part of the picture. How much the picture holds decides how many are heard, and which. Its brightness ties flowers: two at one level — one flips backwards, the other skips. Spora follows the picture closely: brightness its pitch, motion its steps, detail its sweep.



Pad on deep: the plate turns to dusk and a held, tonal sound blooms in orange.

Pad

When a held, tonal sound plays, the pad lays itself over it — long soft grains of it, an octave lower, sunk in a deep room — and keeps it sounding while it lasts (two at most). The `pad` key cycles *off* · *shallow* (one pad at a time) · *middle* · *deep*; deep turns the plate to dusk.

The *orange sun* at the bottom left is the band where a pad may be found: drag its upper half for the top of the band, its lower half for the bottom (60 Hz – 2.4 kHz). Two taps restore it. It is faint

while the pad is off.

06 · THE BAR

The bar

play / stop	Start and stop. The arrow beside it goes back to the start.
mic	Records into 1, 2, 3 … 8 in turn (not while playing).
reset	Forget what it learned: a pad let go, held effects ended, a room's sound no longer judged for a while. Hold it to fire a laser now.
link	Off · on (red): every flower tied at random, again every half minute · on (blue): each flower's onsets — what comes in, not what stays — through a filter drawn at random each time, now and then sliding.
fly	A butterfly; where it lands, it shakes a flower.
grow	Figures spring up and tie the flowers they touch; their roots spread underneath. Laid over a flower's head, a figure is taken in: the flower grows, and its file breaks a little.
drift	The flowers' settings wander slowly. Tap again: the sun wanders too (amber). Again: the pitches glide too (violet). Again: off.
pad	Off · shallow · middle · deep (see above).
voices	How many are heard at once, 1 – 3.
pace	Tap the beat; hold to switch it on or off.
book	The manual inside the app.

07 · NOW AND THEN

Now and then

Burst

A loud place breaks in, crossfaded.

Laser

Very seldom, a flower rising sharply takes the whole: it gathers itself, then fires — it alone, a violet beam turning round it. Hold a flower (or `reset`) to have one now.

Loops

Loops that go round five times move on along the sound.

Shots

Effects arrive as shots — a chain for a while, then gone: filter, delay, short delay, room, lo-fi, grain, glitch, ring, fold …

Keep

The small rings under the moon and the sun. Empty, a tap keeps the plate as it is — every sound, its settings, the figures, the keys. Kept, a tap brings it back. Hold to forget it.

Take

The red ring at the bottom right. Choose a folder and a name (Files › On My iPhone › Humusmus); everything you hear is recorded there as WAV. Tap again to stop. The jar shows the time.

08 · BEFORE YOU START

Requirements

Devices	iPhone and iPad, iOS / iPadOS 18 or later. Up to ten sounds are analysed and played at once and the camera can run alongside, so a recent device is recommended (developed and tested on iPhone 13).
Camera	Used only by the moon, to read brightness, motion and detail. Nothing is recorded or sent anywhere.
Microphone	Used only when you record into a flower.
Files	Load your own audio files; recordings made with TAKE are saved as WAV in Files › On My iPhone › Humusmus.
Version	This guide describes version 1.0.

Tips

Start with few sounds and add more. Very different recordings — a voice, a room, a machine, rain — tie together in more surprising ways than similar ones. If it gets too busy, lower `voices` to 1, or press `reset`.

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